

California State University, Fresno

EYE-DHD

Requirements

Frank Bernal, Devon Webb, Jazper Malone, Anush Nazaryan, Miguel Ibarra, & Joshua Bayt

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Professor Herdman

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Introduction

The Psychology Department at Fresno State has been conducting a study utilizing a mixed reality simulation to find a way to diagnose ADHD using objective data, as opposed to subjective data.

The test subjects are in a room consisting of walls painted green and a desk. They put on a Varjo XR-4 headset and a set of headphones. Once the simulation begins, the test subject is inside a classroom and sits through a 20-25 minute lecture on ethics. Eye-DHD will take the data generated from the simulation and generate an animation of the eye movements and dilation to assist the Psychology Department in analyzing the data.

User Requirements

Features: The user will be able to select data on their device for Eye-DHD to process. Once the data is processed, the system generates a visualization that overlays eye-tracking graphics, such as pupil dilation and gaze movements, directly on the existing video. Researchers will be able to use this tool to manage cases. The interface will support side-by-side comparisons of two cases, enabling direct observation of gaze patterns and differences in pupil dilation. Additionally, the tool will include visualization and analysis features, such as time-series graphs of saccades and pupil dilations, which can be exported for research and diagnosis purposes.

Integration Requirements: One integration requirement is the img.ly API for video editing functionality. The project will use [Electron](#), which will simplify deployment and communication between the front-end and back-end.

The data to be processed is stored as CSV files, and will require PostgreSQL for a relational database to hold the data.

User Interface: The expected user interface will be a web-based dashboard that is clean, intuitive, and focused on guiding users through the workflow of uploading, cleaning, analyzing, and reviewing eye-tracking data. From a User experience perspective, the design should emphasize clarity and ease of use, with step-by-step workflows and responsive feedback (status messages and error notifications).

Constraints: The Project must be finished by May 2026. A budget for this project is constrained to what the Fresno State Psychology department is willing to allocate.

Use Cases

To begin with, the goal has always been to assist in the diagnosis of ADHD, and additionally, to find biological markers that indicate such. It was decided that, through the use of simulation and eye tracking that key markers could be found in which a correlation could be found between them and the disorder.

Another use case for this tool involves education regarding what ADHD is, how it is diagnosed, and how it affects individuals who have it. The tool will help destigmatize biased perspectives surrounding mental health through objective data and diagnosis.

The software may also be used for other disorders as a possible future use case. This allows the same framework to be applied toward identifying markers and providing insights into a broader range of conditions.

Priority & Milestones

High Priority: A function that cleans the raw data obtained from the headset. A function that takes the cleaned data and generates an animation of the eyes with the associated readout data.

Low Priority: The user interface. A Video editor to combine video footage from the headset alongside the animation. A comparison tool for the user to look at 2 test cases simultaneously.

Milestones and Deliverables: (1) A function that cleans raw data and leaves what is needed: pupil dilation measurement, iris measurement, frame, and logtime. (2) A function that processes the cleaned data into an animation that displays the eye measurements over time, with the measurements superimposed onto the animation. (3) A user interface where anyone on the research team can open the application and process their data into an animation for analysis. (4) A video editing feature enabling the user to combine the test case headset video with the corresponding animation. (5) A comparison feature that allows the user to view two test cases simultaneously for analysis.